Year 10 Set 3 · GCSE · Lesson 9

Circles: circumference and area

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Today I will learn to...

Goals

1. Find the circumference of a circle

2. Find the area of a circle

Do now: retrieval practice

4 mins

1. Work out 3.14×4

   12.56

2. Work out 6^2

   36

3. What is half of 18?

   9

4. Round 47.26 to 1 decimal place

   47.3

Objective 1

Find the circumference of a circle

Key idea

Learn

!

Circumference is the distance around a circle. $C = \pi \times d$ (diameter). If you are given the radius, double it first: $d = 2r$. Use the π button and round at the end.

Objective 1: Find the circumference of a circle

I do: example 1

I do

Find the circumference of a circle with diameter 10 cm. Give your answer to 1 dp.

1. $C = \pi \times d$
2. $C = \pi \times 10$
3. $= 31.415\ldots$
4. 31.4 cm

I do: example 2

I do

Find the circumference of a circle with radius 4 cm. Give your answer to 1 dp.

1. Radius 4, so diameter = 2 × 4 = 8 cm
2. $C = \pi \times 8$
3. = 25.132…
4. 25.1 cm

I do: example 3

I do

A bike wheel has diameter 70 cm. How far does it travel in one full turn? Give your answer to the nearest cm.

1. One turn = one circumference
2. $C = \pi \times 70$
3. $= 219.91\ldots$
4. 220 cm

Mini whiteboards

We do

1. Circumference, diameter 6 cm (1 dp)

Answer: 18.8 cm

2. Circumference, radius 5 cm (1 dp)

Answer: 31.4 cm

3. Circumference, diameter 15 cm (1 dp)

Answer: 47.1 cm

4. Circumference, radius 2.5 cm (1 dp)

Answer: 15.7 cm

You do: choose your challenge

You do

Bronze

Circumference, diameter 8 cm (1 dp)

Silver

Circumference, radius 7 cm (1 dp)

Gold

A clock face has diameter 30 cm. Find its circumference to 1 dp.

Platinum

A wheel has radius 35 cm. How far does it travel in 10 turns, in metres to 1 dp?

You do: answers

Check

Bronze — 25.1 cm

Silver — 44.0 cm

Gold — 94.2 cm

Platinum — 70π = 219.9 cm per turn, × 10 = 22.0 m

If you finish early

Extra

Bronze

Circumference, diameter 4 cm (1 dp)

12.6 cm

Silver

Circumference, radius 9 cm (1 dp)

56.5 cm

Gold

A circular running track has diameter 60 m. How far is 3 laps, to the nearest m?

$3 \times 60\pi = 565$ m

Platinum

A circle has circumference 31.4 cm. Find its diameter.

$31.4 \div \pi = 10$ cm

Exam question 1

2 marks

A circle has diameter 12 cm. Work out its circumference. Give your answer to 1 decimal place.

Space for working

Mark scheme: π × 12 (M1) = 37.7 cm (A1)

Exam question 2

2 marks

A circular pond has radius 3.5 m. Work out its circumference, in terms of π.

Space for working

Mark scheme: d = 7 (M1). 7π m (A1)

Objective 1 achieved

Find the circumference of a circle



Objective 2

Find the area of a circle

✓ Achieved: Find the circumference of a circle

Key idea

Learn

!

Area = $\pi \times r^2$. Square the RADIUS first, then multiply by π. If you are given the diameter, halve it first. Area is in square units, e.g. cm^2.

Objective 2: Find the area of a circle

I do: example 1

I do

Find the area of a circle with radius 5 cm. Give your answer to 1 dp.

1. $A = \pi \times r^2$
2. $r^2 = 5 \times 5 = 25$
3. $A = \pi \times 25 = 78.539...$
4. $78.5\ cm^2$

I do: example 2

I do

Find the area of a circle with diameter 12 cm. Give your answer to 1 dp.

1. Diameter 12, so radius = 6 cm
2. $r^2 = 36$
3. $A = \pi \times 36 = 113.097\ldots$
4. $113.1\ \text{cm}^2$

I do: example 3

I do

Find the area of a semicircle with radius 6 cm. Give your answer to 1 dp.

1. Whole circle: $\pi \times 6^2 = 36\pi$
2. A semicircle is half
3. $36\pi \div 2 = 18\pi = 56.548\ldots$
4. $56.5\ \text{cm}^2$

Mini whiteboards

We do

1. Area, radius 3 cm (1 dp)

Answer: 28.3 cm^2

2. Area, diameter 10 cm (1 dp)

Answer: 78.5 cm^2

3. Area, radius 6 cm (1 dp)

Answer: 113.1 cm^2

4. Area, diameter 8 cm (1 dp)

Answer: 50.3 cm^2

You do: choose your challenge

You do

Bronze

Area, radius 4 cm (1 dp)

Silver

Area, diameter 14 cm (1 dp)

Gold

Area of a semicircle, diameter 10 cm (1 dp)

Platinum

A square of side 10 cm has a circle of diameter 10 cm cut out. Find the area left to 1 dp.

You do: answers

Check

Bronze	$50.3\ cm^2$
Silver	$153.9\ cm^2$
Gold	$39.3\ cm^2$
Platinum	$100 - 25\pi = 21.5\ cm^2$

If you finish early

Extra

Bronze

Area, radius 2 cm (1 dp)

$12.6\ cm^2$

Silver

Area, diameter 16 cm (1 dp)

$201.1\ cm^2$

Gold

Area of a quarter circle, radius 10 cm (1 dp)

$78.5\ cm^2$

Platinum

Which is bigger: a circle of radius 5 cm or a square of side 9 cm? By how much?

Circle $78.5\ cm^2$, square $81\ cm^2$: the square, by $2.5\ cm^2$

Exam question 1

2 marks

A circle has radius 8 cm. Work out its area. Give your answer to 1 decimal place.

Space for working

Mark scheme: $\pi \times 8^2$ (M1) = 201.1 cm^2 (A1)

Exam question 2

2 marks

A circular rug has diameter 2 m. Work out its area in terms of π.

Space for working

Mark scheme: r = 1 (M1). π m² (A1)

Objective 2 achieved

Find the area of a circle



Real life

Apply

A pizza has diameter 30 cm. A smaller pizza has diameter 20 cm. How much more pizza do you get with the bigger one?

$\pi \times 15^2 - \pi \times 10^2 = 225\pi - 100\pi = 125\pi \approx 392.7$ cm² more

Extension: stretch yourself

Stretch

1. A circle has circumference 50 cm. Find its radius to 1 dp.

$d = 50 \div \pi = 15.92$, $r = 8.0$ cm

2. A circle has area 100 cm^2. Find its radius to 1 dp.

$r^2 = 100 \div \pi = 31.83$, $r = 5.6$ cm

Exit ticket: exam questions

Exit

1. Find the circumference of a circle with diameter 9 cm (1 dp). **(2 marks)**

2. Find the area of a circle with radius 7 cm (1 dp). **(2 marks)**

3. Find the area of a circle with diameter 20 cm, in terms of π. **(1 mark)**

Exit ticket: mark scheme

Mark it

Q1 $\pi \times 9$ (M1) = 28.3 cm (A1) [2]

Q2 $\pi \times 49$ (M1) = 153.9 cm^2 (A1) [2]

Q3 100π cm^2 [1]

Total: ____ / 5

Today I achieved

- ✔ **Find the circumference of a circle**
- ✔ **Find the area of a circle**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Find the circumference of a circle with diameter 10 cm (1 dp). [2]
2. Find the circumference of a circle with radius 4 cm (1 dp). [2]
3. Find the area of a circle with radius 5 cm (1 dp). [2]
4. Find the area of a circle with diameter 12 cm (1 dp). [2]
5. Find the area of a semicircle with radius 6 cm (1 dp). [2]

Extra practice: Maths Genie → Edexcel question bank → Circles: area and circumference